UNIT 2: METABOLISM & ENERGY

Biology 9 Mr. Queenan

Question

- Why do we produce heat when it is hot (85°F) outside, even though our body temperature is 98.6°F?

- Our bodies are always producing heat.
- At these higher temperatures, we are producing more heat than we need to maintain a core body temperature 98.6°F.
- Thus, we sweat and behave in ways that help release our extra heat generated in cellular respiration.

Energy

- Cells transform energy as they perform work
  - Cells are miniature chemical factories, housing thousands of chemical reactions.
  - Some of these chemical reactions release energy, and others require energy.

Energy

- What is energy?
  - **Energy** is the capacity to cause change or to perform work.
    - Allow living things to carry on processes of life, such as growth, development, metabolism, and reproduction
  - There are two basic forms of energy.

1. **Kinetic energy** is the energy of motion.
2. **Potential energy** is energy that matter possesses as a result of its location or structure.

Energy

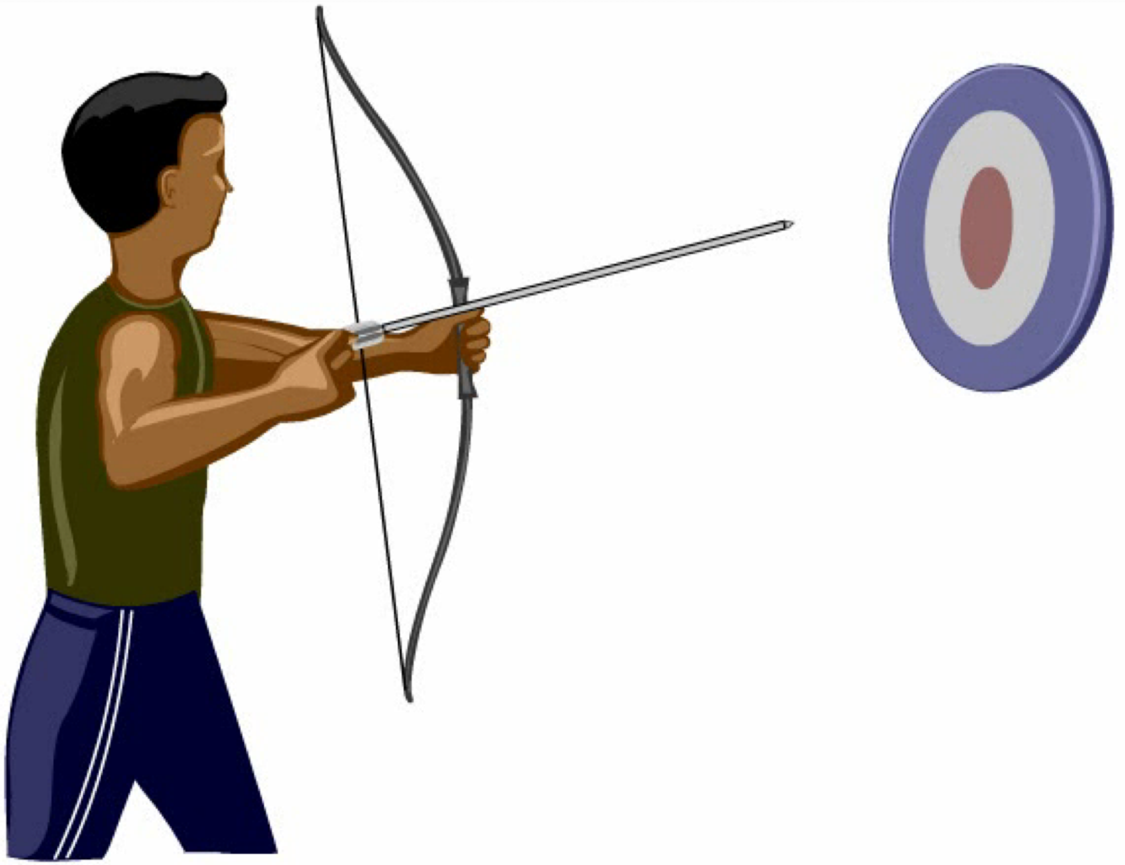
- **Thermal energy** is a type of kinetic energy associated with the random movement of atoms or molecules.
- Thermal energy in transfer from one object to another is called **heat.**
- Light is also a type of kinetic energy; it can be harnessed to power photosynthesis.

https://www.youtube.com/watch?v=LL54E5CzQ-A

Energy

- **Chemical energy** is the
  - potential energy available for release in a chemical reaction and
  - the most important type of energy for living organisms to power the work of the cell.

Energy



Thermodynamics

- **Thermodynamics** is the study of energy transformations that occur in a collection of matter.
  - The word *system* is used for the matter under study.
  - The word *surroundings* is used for everything outside the system; the rest of the universe.

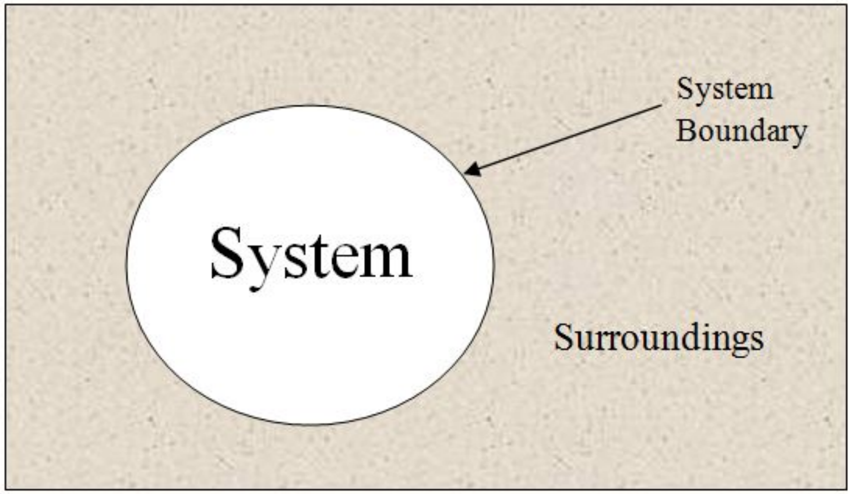
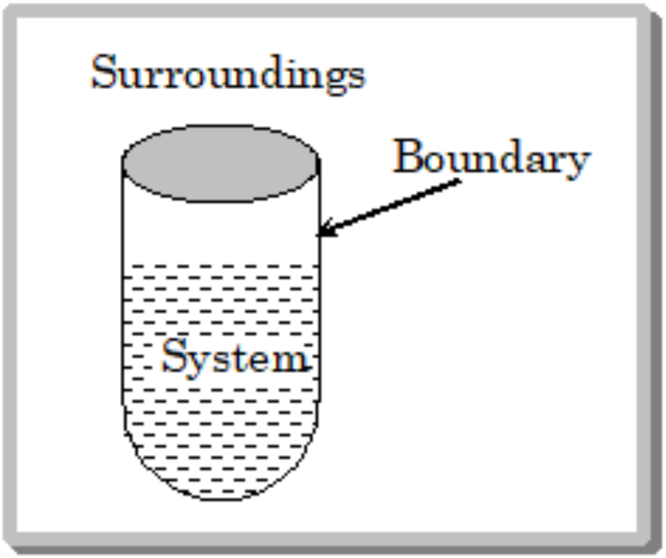


Fig- Thermodynamic system



Thermodynamics

- Two laws govern energy transformations in organisms.
  - Per the **first law of thermodynamics** (also known as the law of energy conservation), energy in the universe is constant.
    - Energy cannot be created or destroyed, only change form
  - Per the **second law of thermodynamics**, energy conversions increase the disorder of the universe.
- **Entropy** is the measure of disorder or randomness.

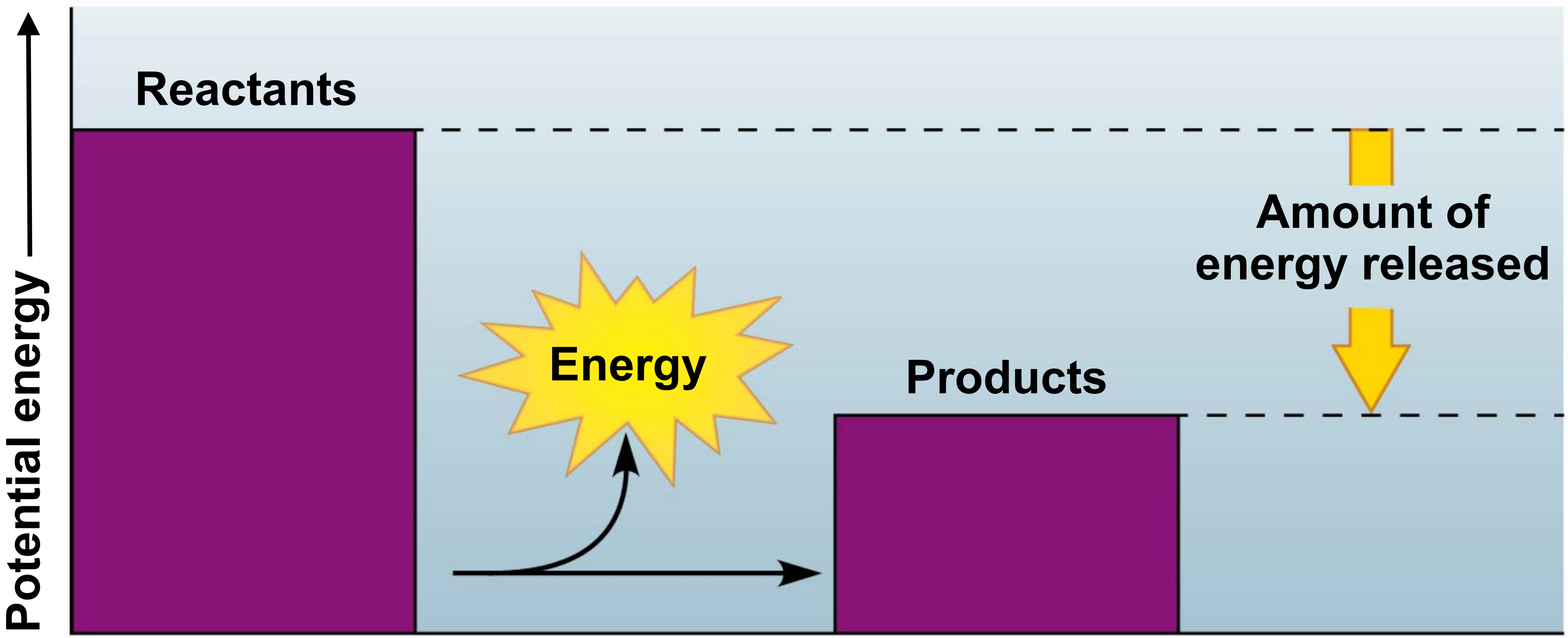
Thermodynamics

- Chemical reactions either
  - release energy (**exergonic reactions**) or
  - require an input of energy and store energy (**endergonic reactions**).

Exergonic Reactions

- **Exergonic reactions** release energy.
  - These reactions release the energy in covalent bonds of the reactants.
  - Examples?
  - Burning wood releases the energy in glucose as heat and light.
  - Cellular respiration
    - involves many steps,
    - releases energy slowly, and
    - uses some of the released energy to produce ATP.

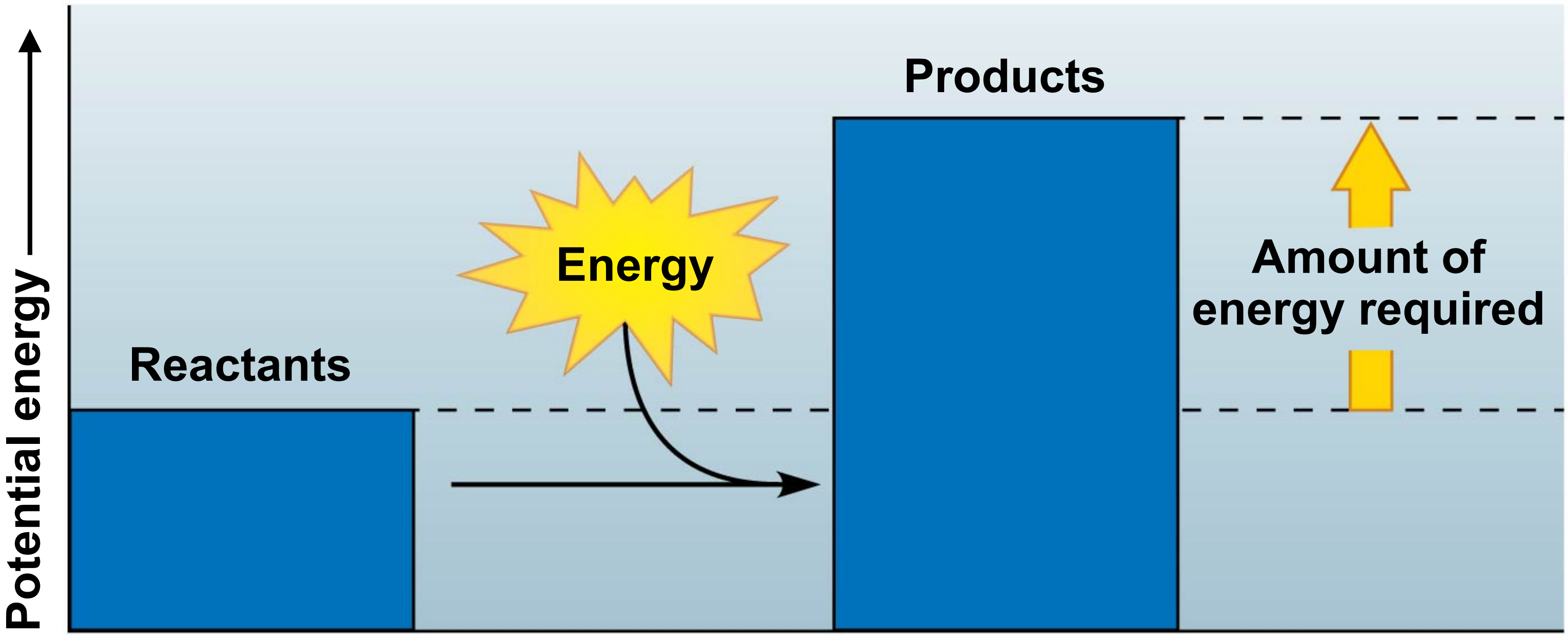
Exergonic Reactions



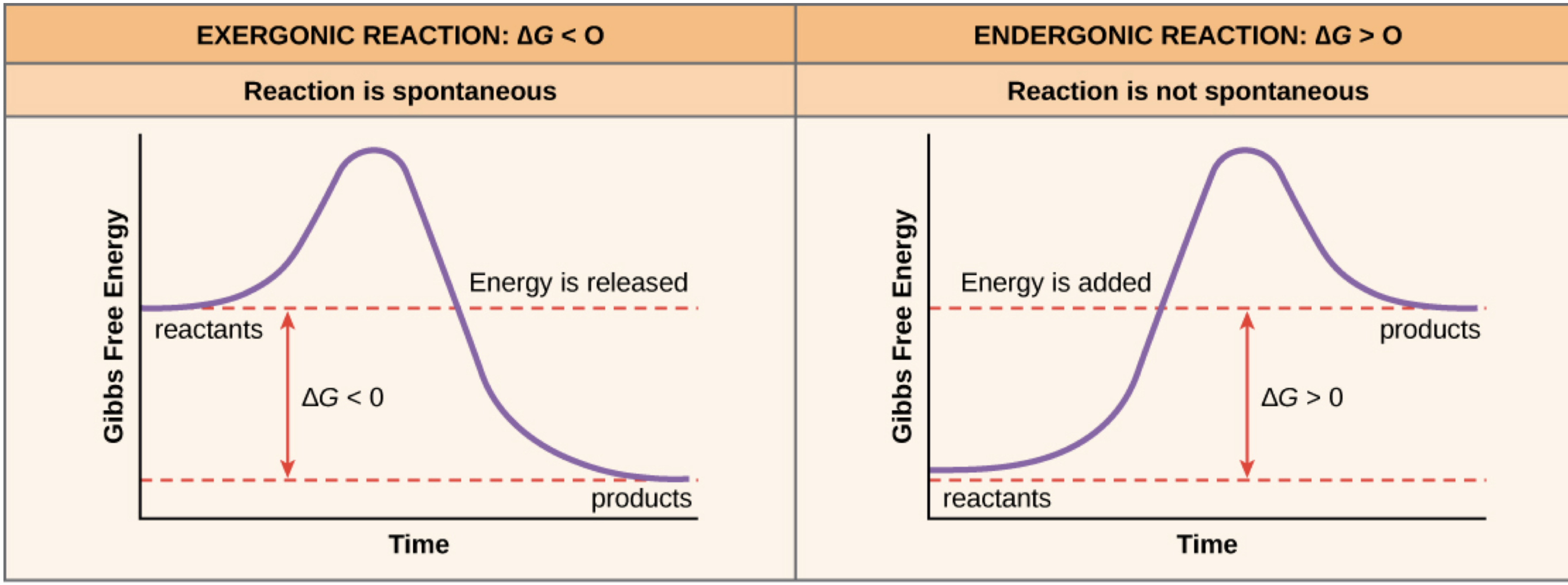
Endergonic Reactions

- An **endergonic reaction**
  - requires an input of energy and
  - yields products rich in potential energy.
- Endergonic reactions
  - start with reactant molecules that contain relatively little potential energy but
  - end with products that contain more chemical energy.
  - Examples include building proteins, DNA, etc. Large molecules with stored energy

Endergonic Reactions



Exergonic v Endergonic



Metabolism

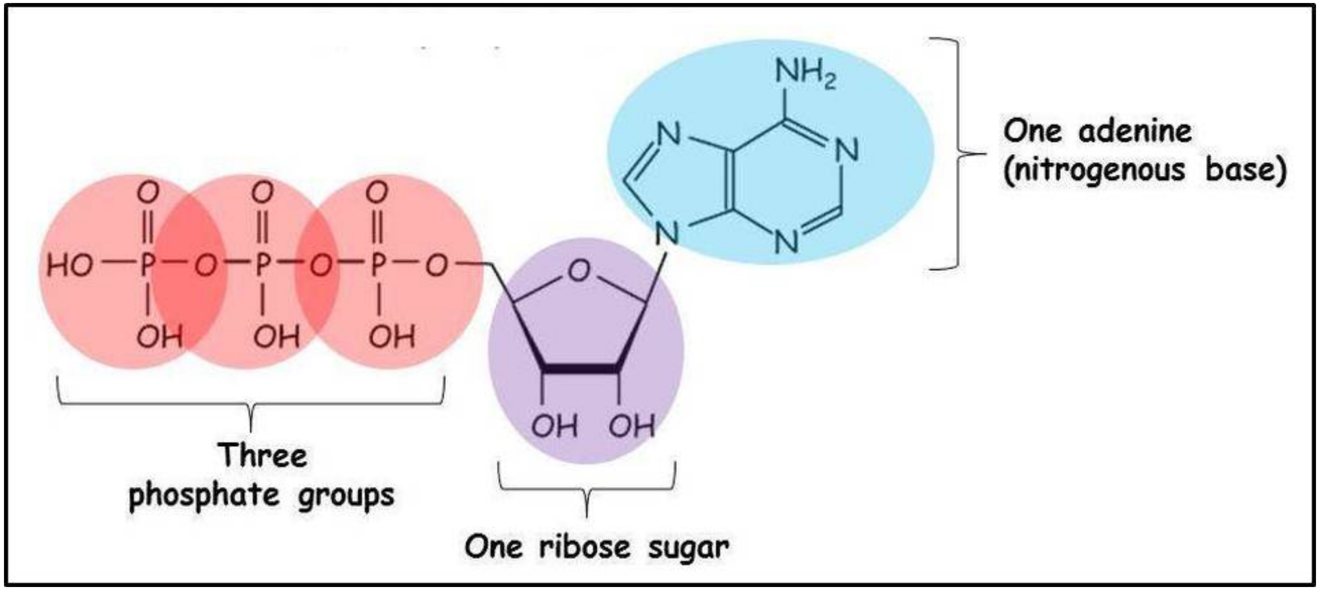
- A living organism carries out thousands of endergonic and exergonic chemical reactions.
- The total of an organism's chemical reactions is called **metabolism.**
- A **metabolic pathway** is a series of chemical reactions that either
  - builds a complex molecule or
  - breaks down a complex molecule into simpler compounds.

ATP

- **Energy coupling** uses the energy released from exergonic reactions to drive endergonic reactions, typically using the energy stored in ATP molecules.

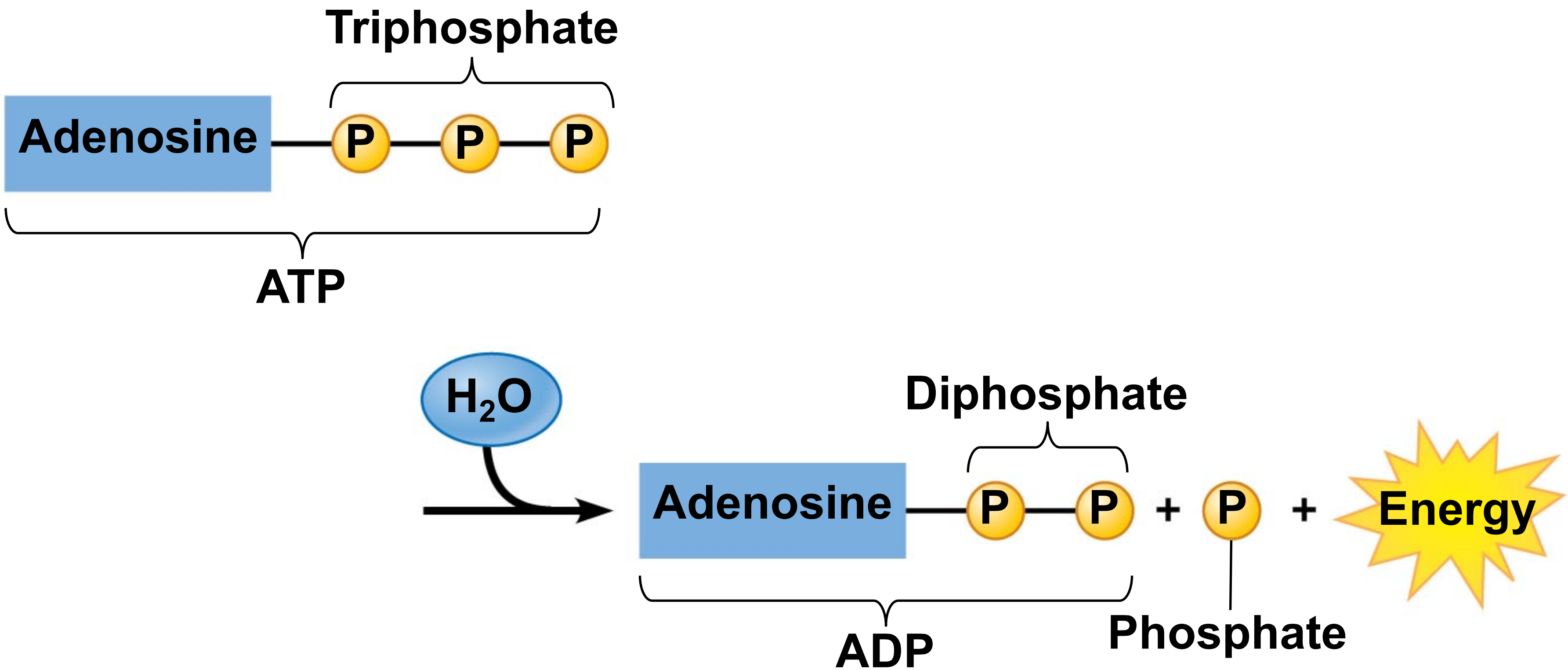
ATP

- **ATP**, adenosine triphosphate, powers nearly all forms of life and consists of
  - adenosine and
  - a triphosphate tail of three phosphate groups.



ATP

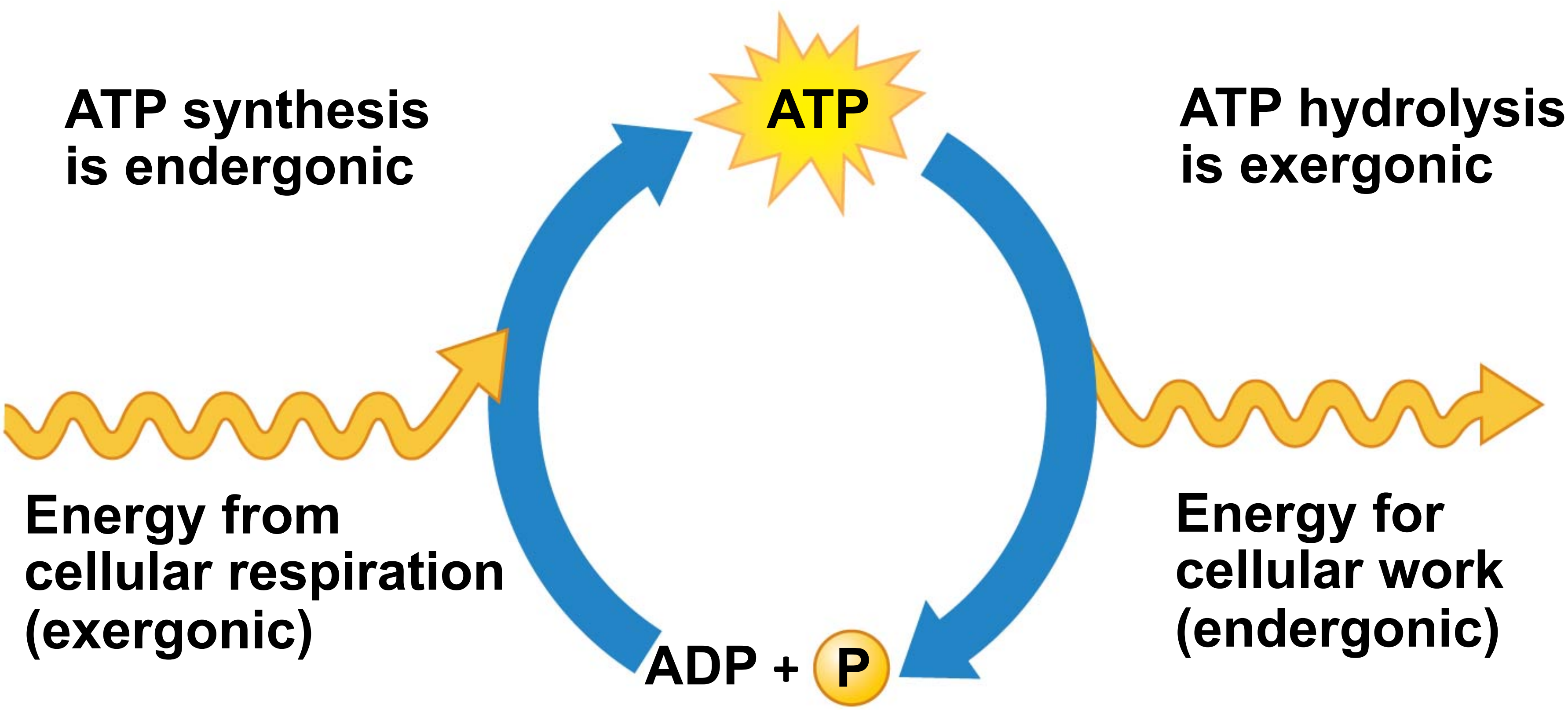
- Hydrolysis of ATP releases energy by transferring its third phosphate from ATP to some other molecule in a process called **phosphorylation.**
- Most cellular work depends on ATP energizing molecules by phosphorylating them.



ATP

- A cell uses and regenerates ATP continuously.
- In the ATP cycle, energy released in an exergonic reaction, such as the breakdown of glucose during cellular respiration, is used in an endergonic reaction to generate ATP from ADP.

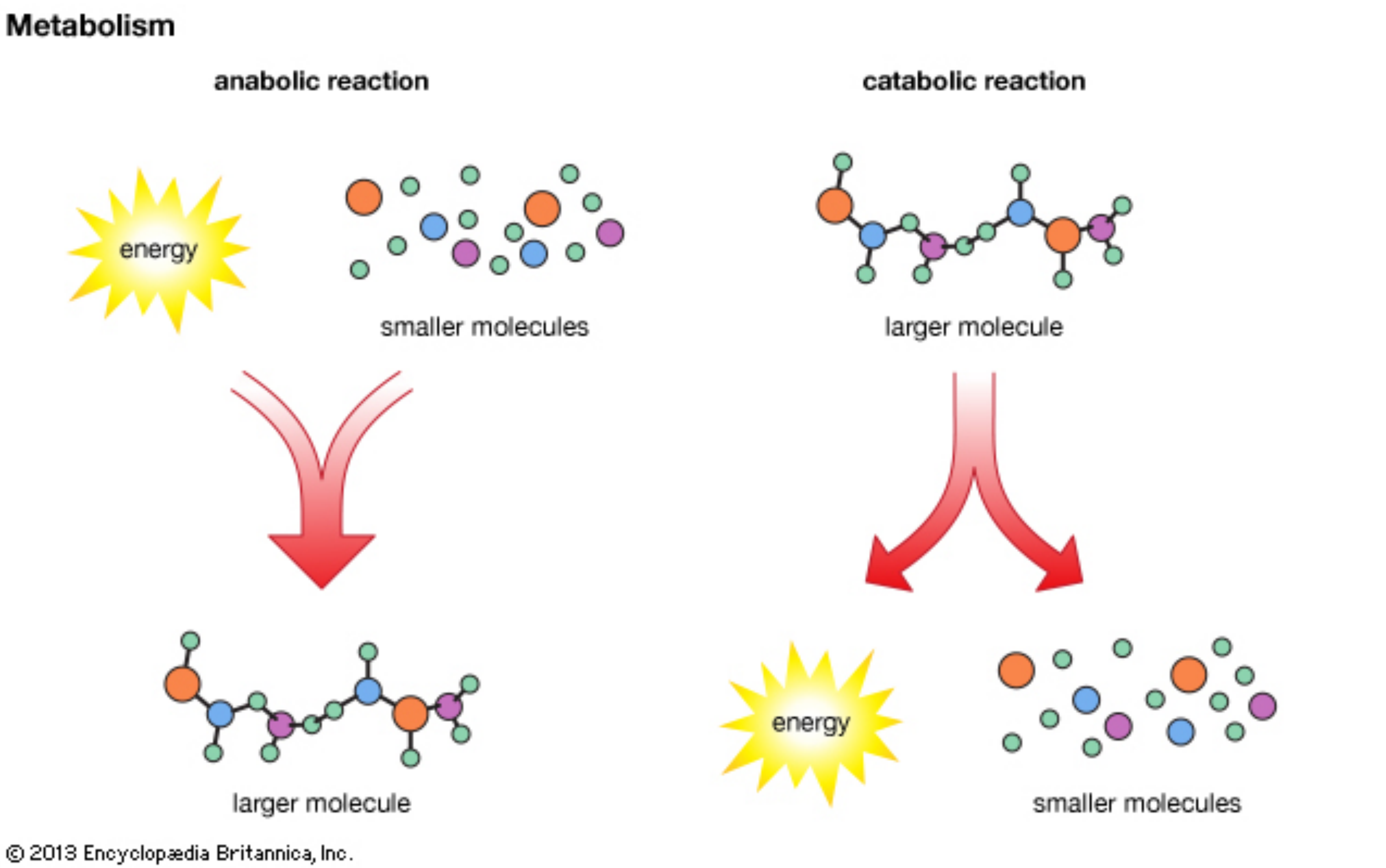
ATP


ATP synthesis
is endergonic
ATP
ATP hydrolysis
is exergonic
Energy from
cellular respiration
(exergonic)
ADP + P
Energy for
cellular work
(endergonic)

Metabolism

- **Catabolism**
  - Relates to the degradative reactions that occur within cells
  - Energy released in the breaking of molecules
    - Proteins --> Amino Acids
- **Anabolism**
  - Relates to the constructive reactions that occur within a cell.
  - Energy must be input in the formation of molecules
    - Glucose --> Complex Carbohydrates

Metabolism


Metabolism
anabolic reaction
catabolic reaction
energy
smaller molecules
larger molecule
larger molecule
energy
smaller molecules
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Metabolism

- Glucose is considered to be the primary source of sugar used to generate ATP
- Fats make excellent cellular fuel because they
  - contain many hydrogen atoms and thus many energy-rich electrons and
  - yield more than twice as much ATP per gram as a gram of carbohydrate.
- Proteins can also be used for fuel, although your body preferentially burns sugars and fats first.

Metabolism

- Metabolism is:
  - the total anabolic and catabolic reactions
  - the total exergonic and endergonic reactions
  - the total energy being used and the total energy being stored